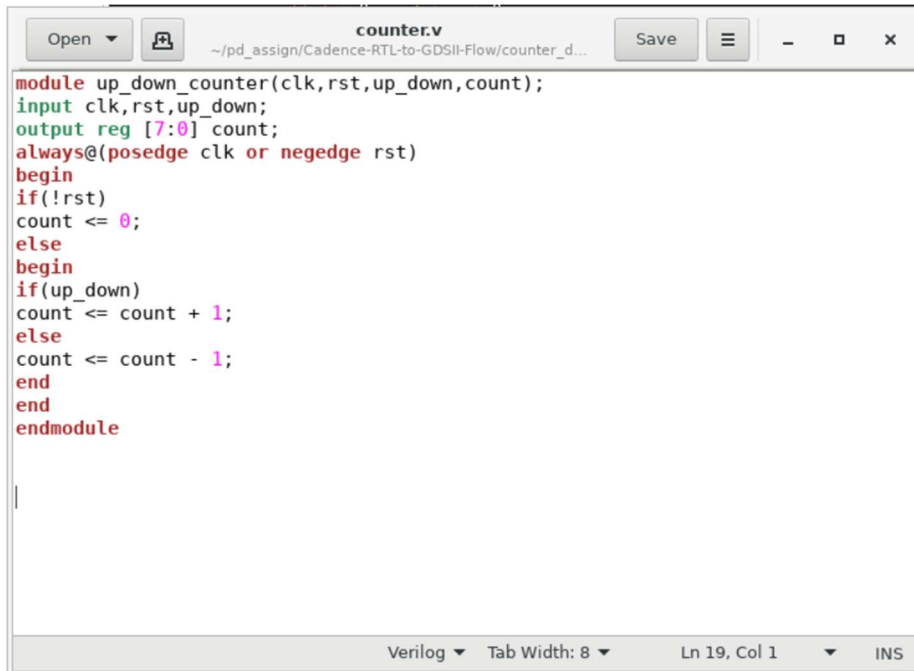


RTL Simulation Part – 1

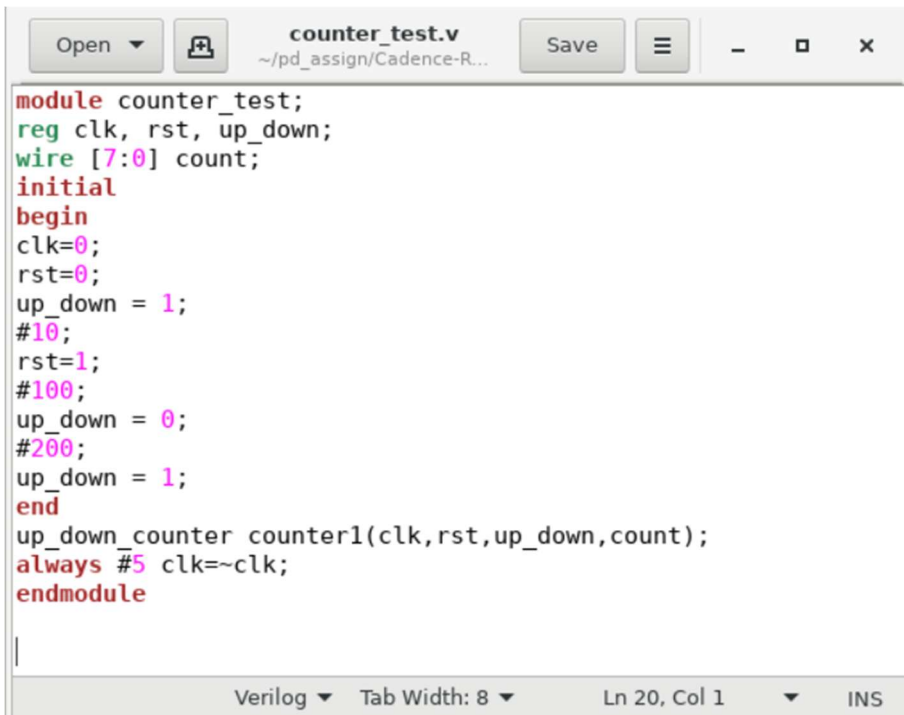
RTL simulation up down counter code:

Design (counter.v) file:



```
module up_down_counter(clk,rst,up_down,count);
input clk,rst,up_down;
output reg [7:0] count;
always@(posedge clk or negedge rst)
begin
if(!rst)
count <= 0;
else
begin
if(up_down)
count <= count + 1;
else
count <= count - 1;
end
end
endmodule
```

Testbench (counter_test.v) file:



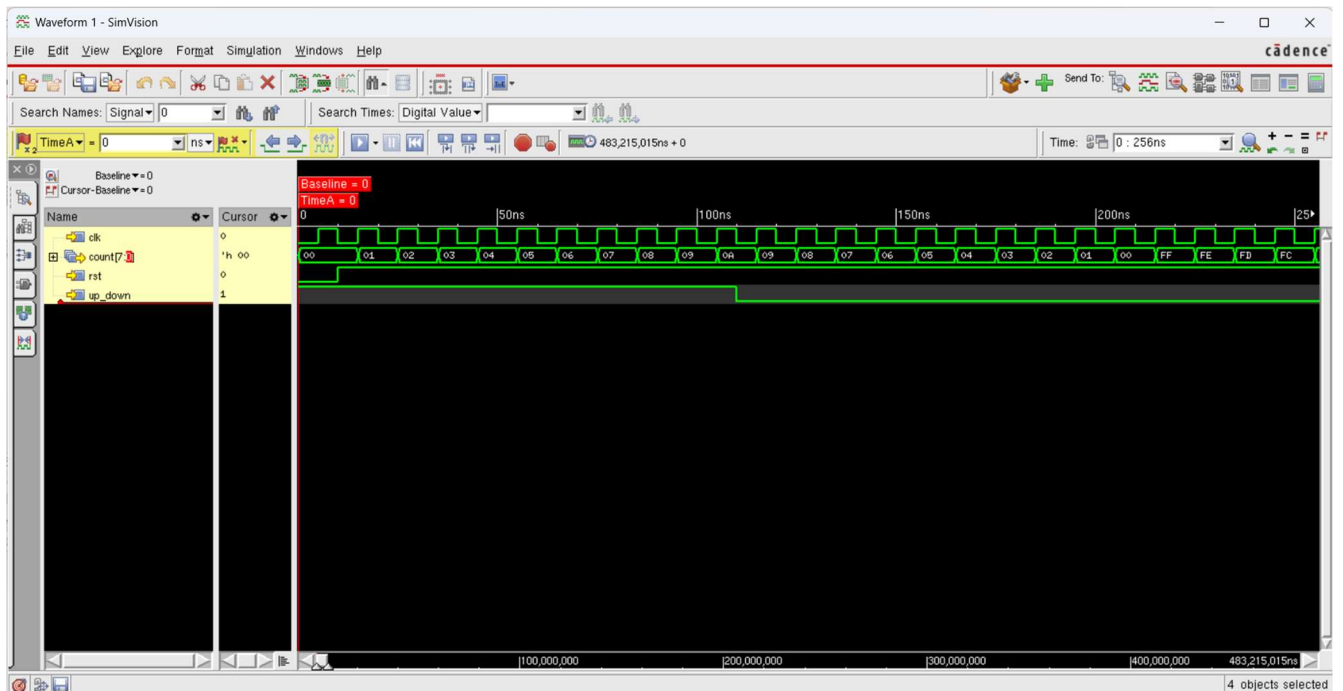
```
module counter_test;
reg clk, rst, up_down;
wire [7:0] count;
initial
begin
clk=0;
rst=0;
up_down = 1;
#10;
rst=1;
#100;
up_down = 0;
#200;
up_down = 1;
end
up_down_counter counter1(clk,rst,up_down,count);
always #5 clk=~clk;
endmodule
```

```

[sg2bt@kc-sse-tux01 simulation]$ irun counter.v counter_test.v -access +rwc -gui &
[1] 2399794
[sg2bt@kc-sse-tux01 simulation]$ T00L: xrun(64)      23.09-s012: Started on Oct 10, 2025 at 12:32:24 CDT
xrun(64): 23.09-s012: (c) Copyright 1995-2024 Cadence Design Systems, Inc.
xrun: *W,NCEXDEP: Executable (irun) is deprecated. Use (xrun) instead.
Recompiling... reason: file './counter_test.v' is newer than expected.
    expected: Mon Jun 19 10:37:16 2017
    actual:   Fri Oct 10 12:32:10 2025
file: counter_test.v
module worklib.counter_test:v
    errors: 0, warnings: 0
    Caching library 'worklib' ..... Done
Elaborating the design hierarchy:
Top level design units:
    counter_test
Building instance overlay tables: ..... Done
Generating native compiled code:
    worklib.up_down_counter:v <0x06a4edab>
        streams: 2, words: 625
    worklib.counter_test:v <0x2693c858>
        streams: 5, words: 2224
Building instance specific data structures.
Loading native compiled code: ..... Done
Design hierarchy summary:
      Instances  Unique
Modules:         2      2
Registers:       4      4
Scalar wires:    3      -
Vectored wires:  1      -
Always blocks:   2      2
Initial blocks:  1      1
Pseudo assignments: 3      -
Process Clocks:  1      1
Writing initial simulation snapshot: worklib.counter_test:v

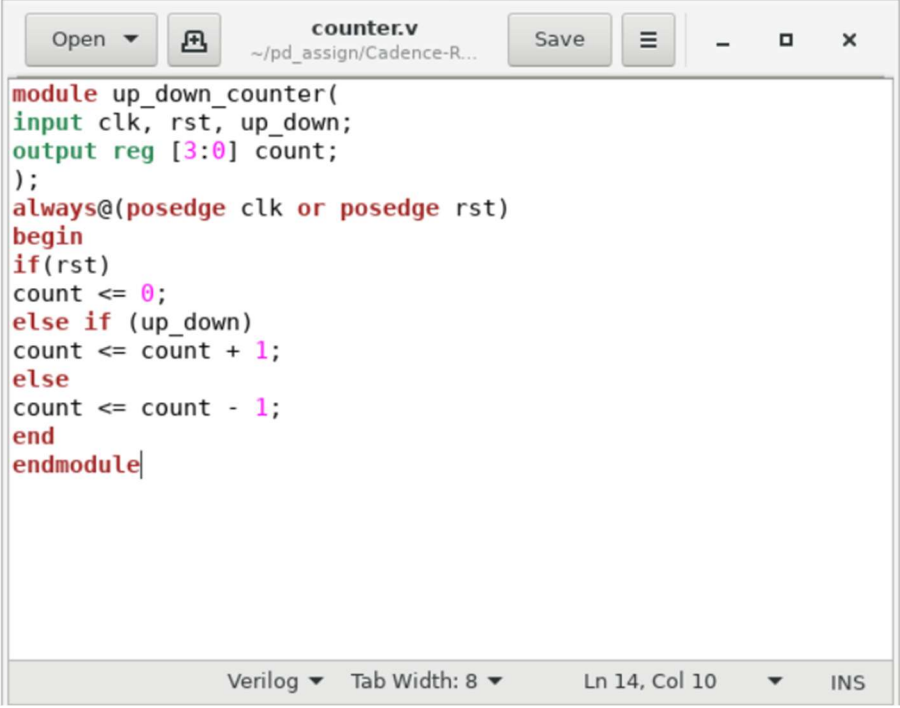
```

Output of Waveforms:



Synthesis Part -2

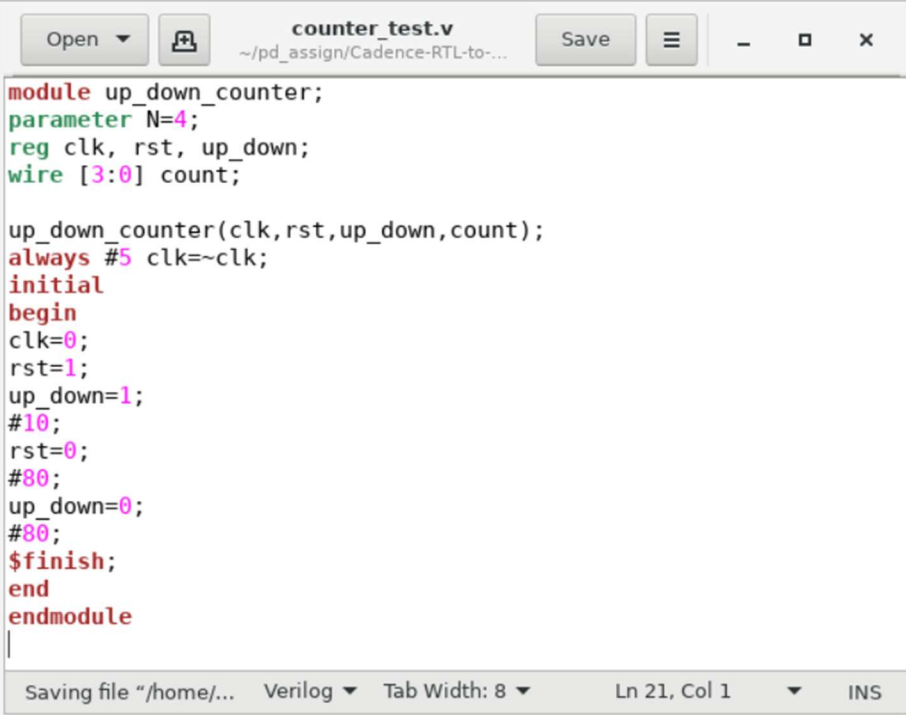
Design code (counter.v file):

A screenshot of a Verilog editor window titled 'counter.v'. The window has a menu bar with 'Open', 'Save', and a hamburger menu icon. The file path is '~/.pd_assign/Cadence-R...'. The code is as follows:

```
module up_down_counter(  
input clk, rst, up_down;  
output reg [3:0] count;  
);  
always@(posedge clk or posedge rst)  
begin  
if(rst)  
count <= 0;  
else if (up_down)  
count <= count + 1;  
else  
count <= count - 1;  
end  
endmodule
```

The status bar at the bottom shows 'Verilog', 'Tab Width: 8', 'Ln 14, Col 10', and 'INS'.

Testbench (counter_test.v file):

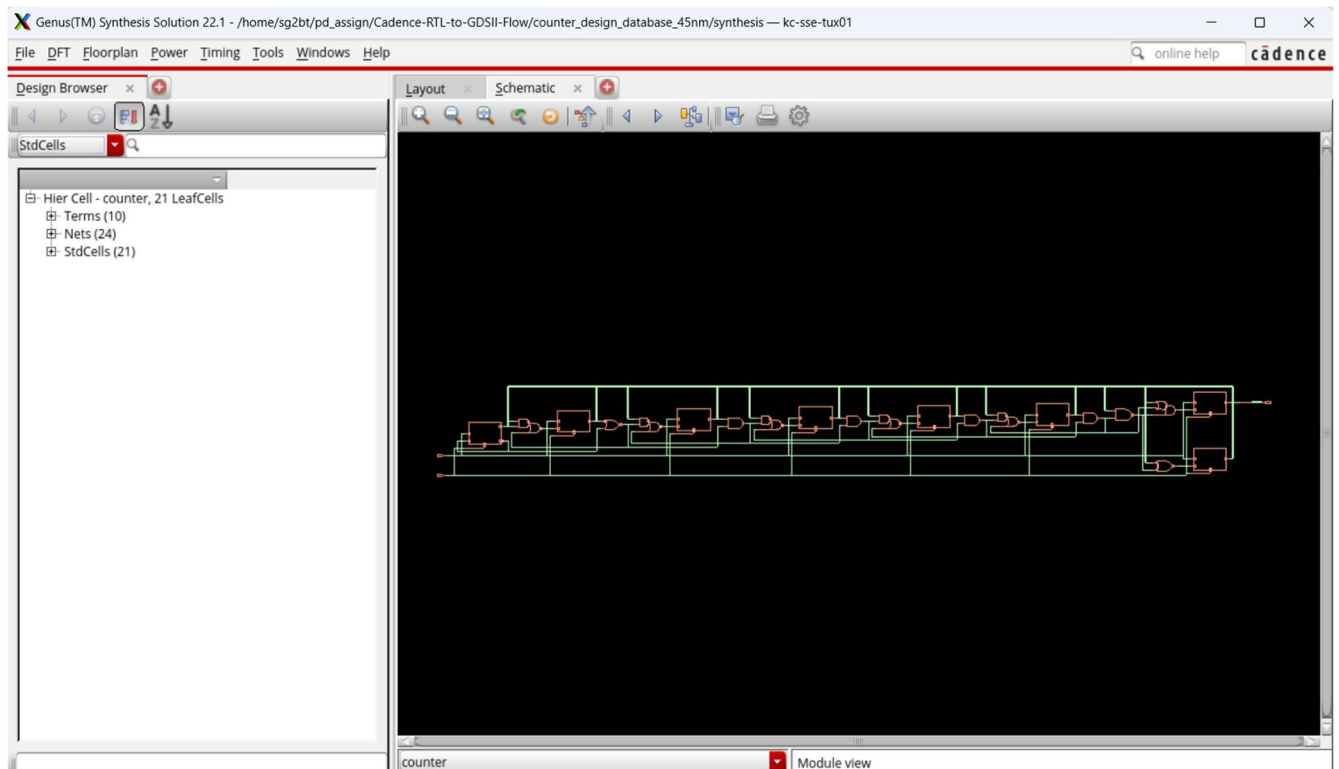
A screenshot of a Verilog editor window titled 'counter_test.v'. The window has a menu bar with 'Open', 'Save', and a hamburger menu icon. The file path is '~/.pd_assign/Cadence-RTL-to-...'. The code is as follows:

```
module up_down_counter;  
parameter N=4;  
reg clk, rst, up_down;  
wire [3:0] count;  
  
up_down_counter(clk,rst,up_down,count);  
always #5 clk=~clk;  
initial  
begin  
clk=0;  
rst=1;  
up_down=1;  
#10;  
rst=0;  
#80;  
up_down=0;  
#80;  
$finish;  
end  
endmodule
```

The status bar at the bottom shows 'Saving file "/>

Schematic view of the design:

Below schematic design of circuit has two input nodes, terms, nets and standard cells.



Generated Netlist:

The below screenshot shows the percentage of area and instance of sequential and logical types.

Generated by: Genus(TM) Synthesis Solution 22.14-s059_1 (Oct 22 2023 08:05:00)
Generated on: Oct 11 2025 13:29:48
Module: design:counter
Technology library: slow_vdd1v0 1.0
Operating conditions: PVT_0P9V_125C (balanced_tree)
Wireload mode: enclosed

Type	Instance	Area	Area_%
Sequential	8	49.932	68.868
Logic	13	22.572	31.132
Total	21	72.504	100

Close Help

Generated Reports:

Below screenshot shows the report of power details for instance, cells in leakage, internal, net and switching values.

Power Details Report — kc-sse-tux01

Generated by: Genus(TM) Synthesis Solution 22.14-s059_1 (Oct 22 2023 08:05:00)
Generated on: Oct 11 2025 13:28:02
Module: design:counter
Technology library: slow_vdd1v0 1.0
Operating conditions: PVT_0P9V_125C (balanced_tree)
Wireload mode: enclosed

Instance	Cells	Leakage (nW)	Internal (nW)	Net (nW)	Switching (nW)
counter	21	1.559	2581.953	189.842	2771.796

Close

Help

Below screenshot shows the report of gate count for instance, cells on cell area, net area, total area.

Gate Count Report — kc-sse-tux01

Generated by: Genus(TM) Synthesis Solution 22.14-s059_1 (Oct 22 2023 08:05:00)
Generated on: Oct 11 2025 13:33:54
Module: design:counter
Technology library: slow_vdd1v0 1.0
Operating conditions: PVT_0P9V_125C (balanced_tree)
Wireload mode: enclosed

Instance	Cells	Cell Area	Net Area	Total Area	Wireload	WL_Flag
counter	21	72.504	0.000	72.504		(S)

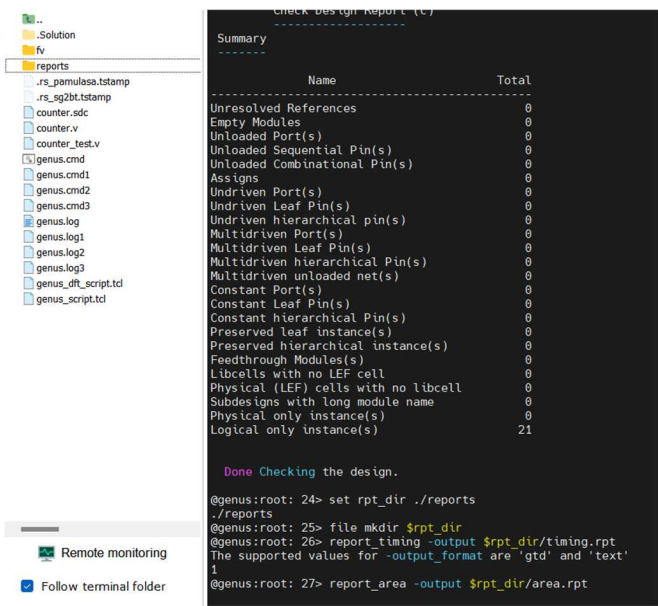
☐ Filtering

Close

Help

Generated Reports Directory:

Below screenshot shows the generated reports folder in the left side.



Report timing:

Below shows the timings of report generated in Mobaxterm.

Exceptions/Constraints:

output_delay 1000 constraints_top.sdc_line_6_7_1

#	Timing Point	Flags	Arc	Edge	Cell	Fanout	Load (fF)	Trans (ps)	Delay (ps)	Arrival (ps)	Instance Location
#	count_reg[0]/CK	-	-	R	(arrival)	8	-	100	0	0	(-, -)
#	count_reg[0]/Q	-	CK->Q	R	DFFRX1	2	0.2	24	316	316	(-, -)
#	count[0]	<<<	-	R	(port)	-	-	-	0	316	(-, -)

Report Power:

Below shows the power of report generated in Mobaxterm.

Instance: /counter

Power Unit: W

PDB Frames: /stim#0/frame#0

Category	Leakage	Internal	Switching	Total	Row%
memory	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
register	1.08261e-09	2.39359e-06	2.40542e-08	2.41872e-06	87.21%
latch	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
logic	4.76223e-10	1.88368e-07	3.61881e-08	2.25032e-07	8.11%
bbox	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
clock	0.00000e+00	0.00000e+00	1.29600e-07	1.29600e-07	4.67%
pad	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pm	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
Subtotal	1.55883e-09	2.58195e-06	1.89842e-07	2.77335e-06	99.99%
Percentage	0.06%	93.10%	6.85%	100.00%	100.00%

Report Area:

Below shows the area of report generated in Mobaxterm.

Instance	Module	Cell Count	Cell Area	Net Area	Total Area	Wireload
counter		21	72.504	0.000	72.504	<none> (D)
(D) = wireload is default in technology library						

Report QoR:

Below shows the quality of report generated in Mobaxterm.

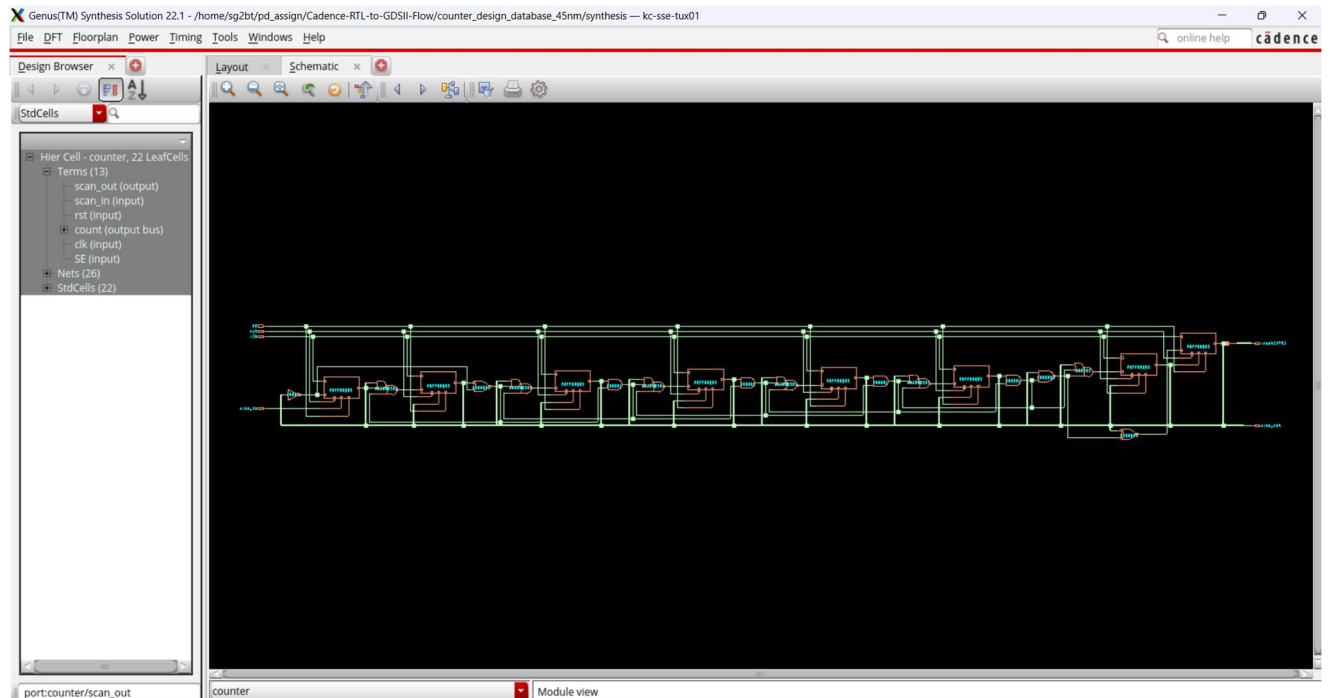
```
Max Fanout          8 (clk)
Min Fanout          0 (rst)
Average Fanout      2.3
Terms to net ratio  3.2500
Terms to instance ratio 3.7143
Runtime             57.911518 seconds
Elapsed Runtime     1130 seconds
Genus peak memory usage 1792.42
Innovus peak memory usage no_value
Hostname            kc-sse-tux01.kc.umkc.edu
```

```
Number of threads: 8 * 1 (id: pbs_debug, time_info v1.57)
Info: (*N*) indicates data that was populated from previously saved time_info database
Info: CPU time includes time of parent + longest thread
##>===== Cadence Confidential (Mapping-Logical) =====
##>Main Thread Summary:
##>-----
##>STEP                Elapsed      WNS      TNS      Insts      Area      Memory
##>-----
##>M:Initial            0          -      -      37         149       884
##>M:Pre Cleanup        0          -      -      37         149       884
##>M:Setup              0          -      -      -          -         -
##>M:Early Clock Gating 0          -      -      -          -         -
##>M:Launch ST          0          -      -      -          -         -
##>M:Design Partition   0          -      -      -          -         -
##>M:Create Partition Netlists 0          -      -      -          -         -
##>M:Init Power         0          -      -      -          -         -
##>M:Budgeting          0          -      -      -          -         -
##>M:Derenv-DB          0          -      -      -          -         -
##>M:Debug Outputs      0          -      -      -          -         -
##>M:ST loading         0          -      -      -          -         -
##>M:Distributed        0          -      -      -          -         -
##>M:Timer              0          -      -      -          -         -
##>M:Assembly           0          -      -      -          -         -
##>M:DFT                0          -      -      -          -         -
##>M:DP Operations      2          -      -      22         72        884
##>M:Const Prop         0          -      0      22         72        884
##>M:Cleanup            0          -      0      22         72        884
##>M:MBCI               0          -      -      22         72        884
##>M:PostGen Opt        0          -      -      -          -         -
##>M:Early Clock Gating Cleanup 0          -      -      -          -         -
##>M:Const Gate Removal 0          -      -      -          -         -
##>M:Misc               1          -      -      -          -         -
##>-----
##>Total Elapsed        3
##>=====
Info      : Done mapping. [SYNTH-5]
Info      : Done mapping 'counter'.
flow.cputime flow.realtime timing.setup.tns timing.setup.wns snapshot
UM:*      syn_map
```

PART-3

Running Synthesis with DFT for generating Netlist in Up down Counter:

Schematic view of the design:



Screenshot of percentage of total registers that are scannable:

```
@genus:root: 8> set db dft prefix dft_
Setting attribute of root '/': 'dft_prefix' = dft_
1 dft_
@genus:root: 9> define_shift_enable -name SE -active high -create_port SE
test_signal:counter/SE
@genus:root: 10> check_dft_rules
Checking DFT rules for 'counter' module under 'muxed_scan' style

Detected 0 DFT rule violation(s)
Summary of check_dft_rules
*****
Number of usable scan cells: 48
Clock Rule Violations:
-----
Internally driven clock net: 0
Tied constant clock net: 0
Undriven clock net: 0
Conflicting async & clock net: 0
Misc. clock net: 0

Async. set/reset Rule Violations:
-----
Internally driven async net: 0
Tied active async net: 0
Undriven async net: 0
Misc. async net: 0

Total number of DFT violations: 0

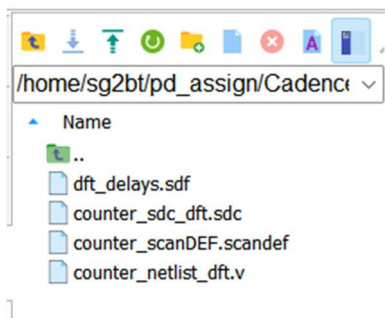
Total number of Test Clock Domains: 1
Number of user specified non-Scan registers: 0
Number of registers that fail DFT rules: 0
Number of registers that pass DFT rules: 8
Percentage of total registers that are scannable: 100%
0
@genus:root: 11>
```

Scan chains reported:

```
@genus:root: 28> report_scan_chains
Reporting 1 scan chain (muxed_scan)

Chain 1: top_chain
scan_in:      scan_in
scan_out:     scan_out
shift_enable: SE (active high)
clock_domain: clk (edge: rise)
length: 8
bit 1        count_reg[0] <clk (rise)>
bit 2        count_reg[1] <clk (rise)>
bit 3        count_reg[2] <clk (rise)>
bit 4        count_reg[3] <clk (rise)>
bit 5        count_reg[4] <clk (rise)>
bit 6        count_reg[5] <clk (rise)>
bit 7        count_reg[6] <clk (rise)>
bit 8        count_reg[7] <clk (rise)>
-----
```

Generated reports directory:



Generated Netlist file:

```
counter_netlist_dft.v
5 // Verification Directory fv/counter
6
7 module counter(clk, rst, count, SE, scan_in, scan_out);
8   input clk, rst, SE, scan_in;
9   output [7:0] count;
10  output scan_out;
11  wire clk, rst, SE, scan_in;
12  wire [7:0] count;
13  wire scan_out;
14  wire n_0, n_1, n_2, n_3, n_4, n_5, n_6, n_7;
15  wire n_8, n_9, n_10, n_11, n_12, n_13;
16  assign scan_out = count[7];
17  SDFFRHQX1 \count_reg[7] (.RN (rst), .CK (clk), .D (n_13), .SI
18    (count[6]), .SE (SE), .Q (count[7]));
19  SDFFRHQX1 \count_reg[6] (.RN (rst), .CK (clk), .D (n_12), .SI
20    (count[5]), .SE (SE), .Q (count[6]));
21  SDFFRHQX1 \count_reg[5] (.RN (rst), .CK (clk), .D (n_9), .SI
22    (count[4]), .SE (SE), .Q (count[5]));
23  XNOR2X1 g184_2398(.A (count[7]), .B (n_11), .Y (n_13));
24  OA21X1 g186_5107(.A0 (count[6]), .A1 (n_10), .B0 (n_11), .Y (n_12));
25  SDFFRHQX1 \count_reg[4] (.RN (rst), .CK (clk), .D (n_8), .SI
26    (count[3]), .SE (SE), .Q (count[4]));
27  NAND2X1 g188_6260(.A (count[6]), .B (n_10), .Y (n_11));
28  AOI2BB1X1 g189_4319(.A0N (count[5]), .A1N (n_7), .B0 (n_10), .Y
29    (n_9));
30  SDFFRHQX1 \count_reg[3] (.RN (rst), .CK (clk), .D (n_5), .SI
31    (count[2]), .SE (SE), .Q (count[3]));
32  SDFFRHQX1 \count_reg[2] (.RN (rst), .CK (clk), .D (n_3), .SI
33    (count[1]), .SE (SE), .Q (count[2]));
34  AOI2BB1X1 g192_8428(.A0N (count[4]), .A1N (n_6), .B0 (n_7), .Y
35    (n_8));
36  AND2XL g191_5526(.A (count[5]), .B (n_7), .Y (n_10));
37  AND2XL g194_6783(.A (count[4]), .B (n_6), .Y (n_7));
38  AOI2BB1XL g195_3680(.A0N (count[3]), .A1N (n_4), .B0 (n_6), .Y
39    (n_5));
40  SDFFRHQX1 \count_reg[1] (.RN (rst), .CK (clk), .D (n_2), .SI
41    (count[0]), .SE (SE), .Q (count[1]));
42  AND2XL g197_1617(.A (count[3]), .B (n_4), .Y (n_6));
43  AOI2BB1X1 g198_2802(.A0N (count[2]), .A1N (n_1), .B0 (n_4), .Y
44    (n_3));
45  SDFFRHQX1 \count_reg[0] (.RN (rst), .CK (clk), .D (n_0), .SI
46    (scan_in), .SE (SE), .Q (count[0]));
47  AOI2BB1X1 g201_1705(.A0N (count[1]), .A1N (count[0]), .B0 (n_1), .Y
48    (n_2));
49  AND2XL g200_5122(.A (count[2]), .B (n_1), .Y (n_4));
50  NOR2BX1 g203_8246(.AN (count[1]), .B (n_0), .Y (n_1));
51  INVX1 g204(.A (count[0]), .Y (n_0));
52 endmodule
53
54
```

PART-4

Logic Equivalence Check for revised Netlist:

Compare Revised vs Golden Netlist Files:

DesignLibraryTool SetupDesign DataRule Checking

Golden

Revised

counter

1 library cells and 9 primitives

U\$1(INV)

add_9(VDW_ADD_8_1_0_CIM)

count_reg[0](DFF)

count_reg[1](DFF)

count_reg[2](DFF)

count_reg[3](DFF)

count_reg[4](DFF)

count_reg[5](DFF)

count_reg[6](DFF)

count_reg[7](DFF)

counter

22 library cells and 1 primitives

U\$1(WIRE)

count_reg[0](SDDFRHQX1)

count_reg[1](SDDFRHQX1)

count_reg[2](SDDFRHQX1)

count_reg[3](SDDFRHQX1)

count_reg[4](SDDFRHQX1)

count_reg[5](SDDFRHQX1)

count_reg[6](SDDFRHQX1)

count_reg[7](SDDFRHQX1)

g184_2398(XNOR2X1)

g186_5107(OA21X1)

g188_6260(NAND2X1)

g189_4319(AOI2BB1X1)

g191_5526(AND2XL)

g192_8428(AOI2BB1X1)

g194_6783(AND2XL)

g195_3680(AOI2BB1X1)

g197_1617(AND2XL)

g198_2802(AOI2BB1X1)

g200_5122(AND2XL)

g201_1705(AOI2BB1X1)

g203_8246(NOR2BX1)

g204(INVX1)

DesignLibraryTool SetupDesign DataRule Checking

Golden

Revised

work_lib (548)

ACHCONX2

ADDFHX1

ADDFHX2

ADDFHX4

ADDFHXL

ADDFX1

ADDFX2

ADDFX4

ADDFXL

ADDXH1

ADDXH2

ADDXH4

ADDXHL

AND2X1

AND2X2

AND2X4

AND2X6

AND2X8

AND2XL

AND3X1

AND3X2

AND3X4

AND3X6

AND3X8

AND3XL

AND4X1

AND4X2

AND4X4

AND4X6

work_lib (548)

ACHCONX2

ADDFHX1

ADDFHX2

ADDFHX4

ADDFHXL

ADDFX1

ADDFX2

ADDFX4

ADDFXL

ADDXH1

ADDXH2

ADDXH4

ADDXHL

AND2X1

AND2X2

AND2X4

AND2X6

AND2X8

AND2XL

AND3X1

AND3X2

AND3X4

AND3X6

AND3X8

AND3XL

AND4X1

AND4X2

AND4X4

AND4X6

File List (Golden):

../rtl/counter.v

File List (Revised):

../synthesis/outputs/counter_netlist_dft.v

Design Data Summary

Design	Golden	Revised
Design-Modules	1	1
Library-Cells	1	22
Primitives		
INPUT	2	4
OUTPUT	8	9
AND	6	26
DFF	8	8
INV	2	28
NAND	0	1
NOR	0	6
OR	0	9
WIRE	0	1
XNOR	0	1
XOR	7	0

Verification Report:

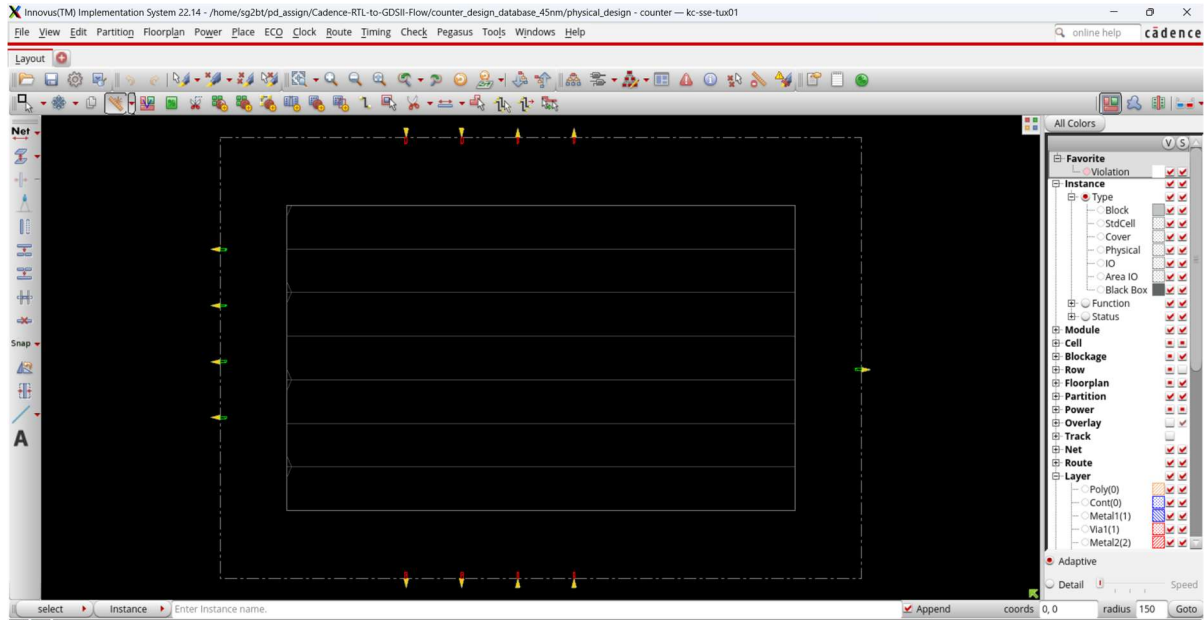
```
--
12 // Command: report verification
13 =====
14                               Verification Report
15 -----
16 Category                                                                Count
17 -----
18 1. Non-standard modeling options used:                                0
19 -----
20 2. Incomplete verification:                                           1
21     Command "add ignore outputs" used:                               yes
22 -----
23 3. User modification to design:                                       0
24 -----
25 4. Conformal Constraint Designer clock domain crossing checks recommended: 0
26 -----
27 5. Design ambiguity:                                                  0
28 -----
29 6. Compare Results:                                                  PASS
30 =====

33 // Note: Read VERILOG design successfully
34 // Command: read design ../synthesis/outputs/counter_netlist_dft.v -verilog -revised
35 // Parsing file ../synthesis/outputs/counter_netlist_dft.v ...
36 // Revised root module is set to 'counter'
37 // Note: Read VERILOG design successfully
38 // Command: add pin constraints 0 SE -revised
39 // Command: add ignored inputs scan_in -revised
40 // Command: add ignored outputs scan_out -revised
41 // Command: set system mode lec
42 // Processing Golden ...
43 // Modeling Golden ...
44 // Processing Revised ...
45 // Modeling Revised ...
46 // Warning: Golden and Revised have different numbers of key points:
47 // Golden key points = 18
48 // Revised key points = 19
49 // Mapping key points ...
50 =====
51 Mapped points: SYSTEM class
52 -----
53 Mapped points      PI      PO      DFF      Total
54 -----
55 Golden             2       8       8       18
56 -----
57 Revised            2       8       8       18
58 =====
59 Unmapped points:
60 =====
61 Revised:
62 -----
63 Unmapped points    PI      Total
64 -----
65 Extra              1       1
66 =====
67 // Command: add compared point -all
68 // 16 compared points added to compare list
69 // Command: compare
70 =====
71 Compared points    PO      DFF      Total
72 -----
73 Equivalent         8       8       16
74 =====
```

PART-5

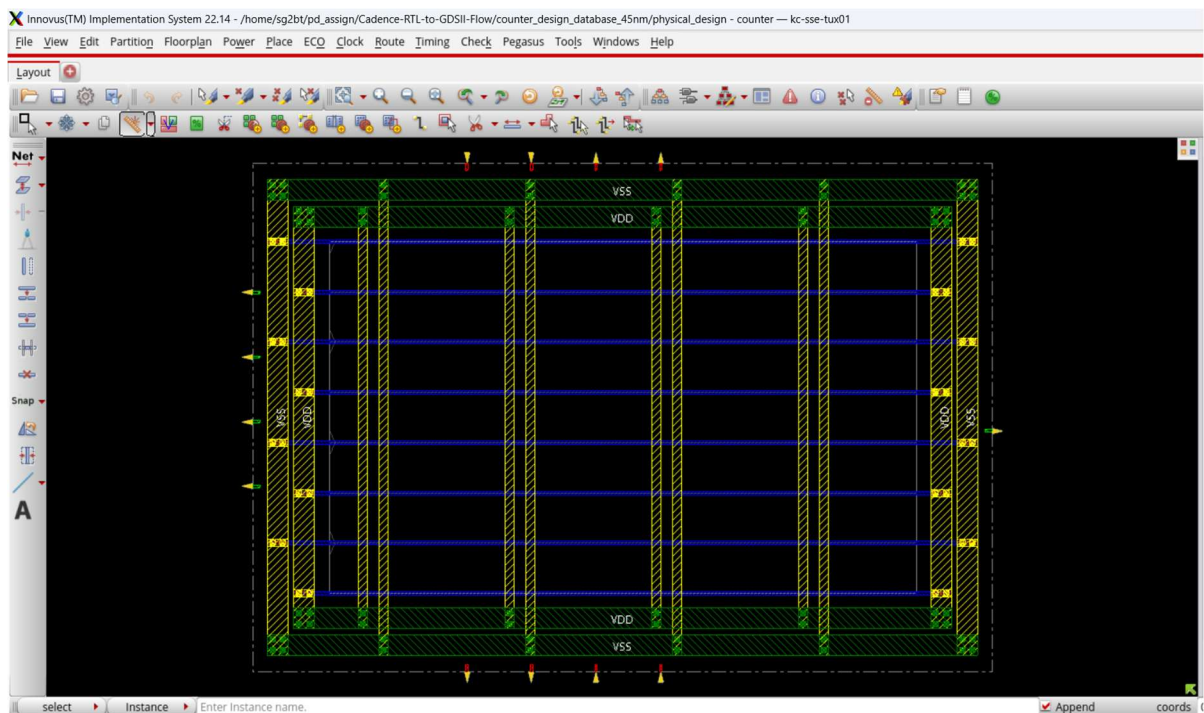
Floorplan Lab:

Below is the screenshot for floor planning stage that showing the IO pins on the boundary cell.



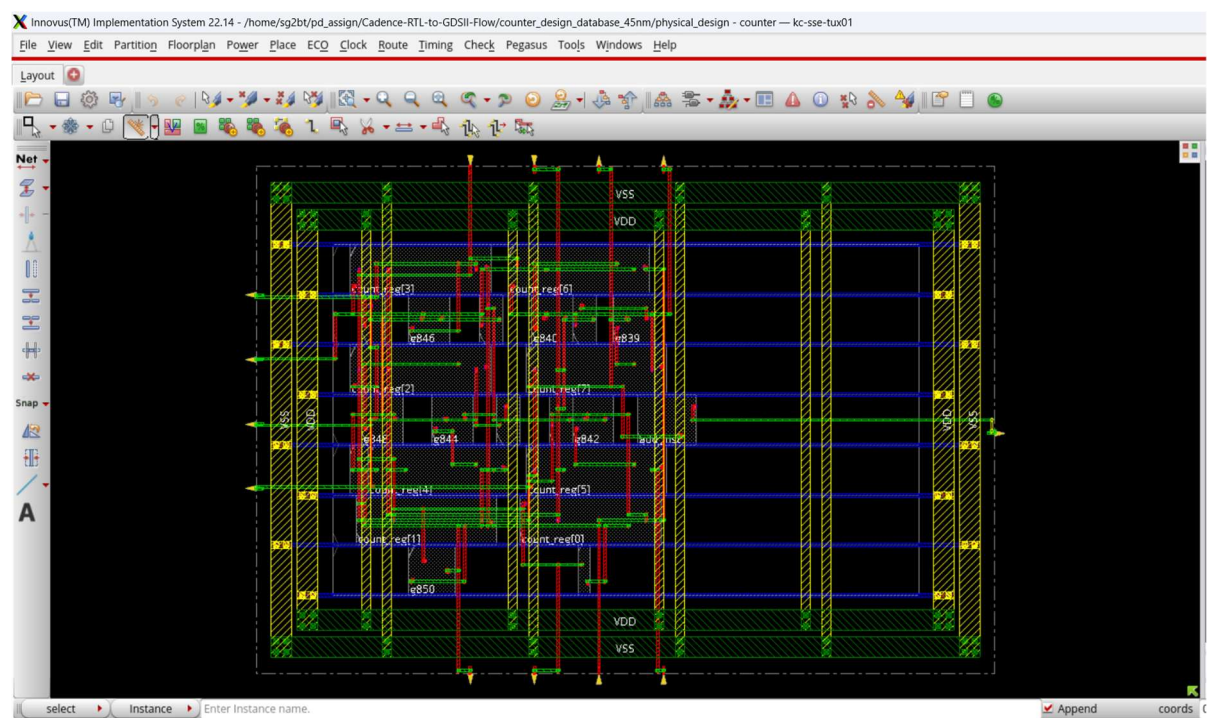
Power plan Lab:

Below is the screenshot for power planning stage that showing the VDD and VSS grid lines.



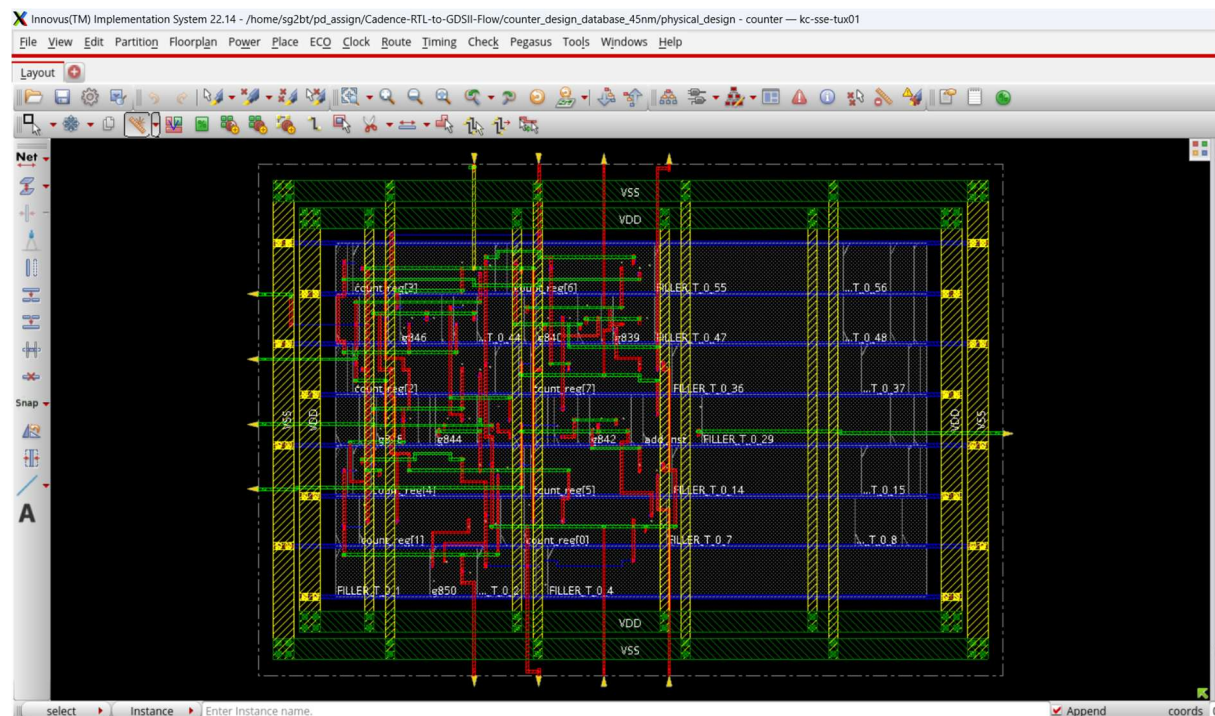
Placement Lab:

Below is the screenshot for Placement stage that showing the connections for up down counter in it.



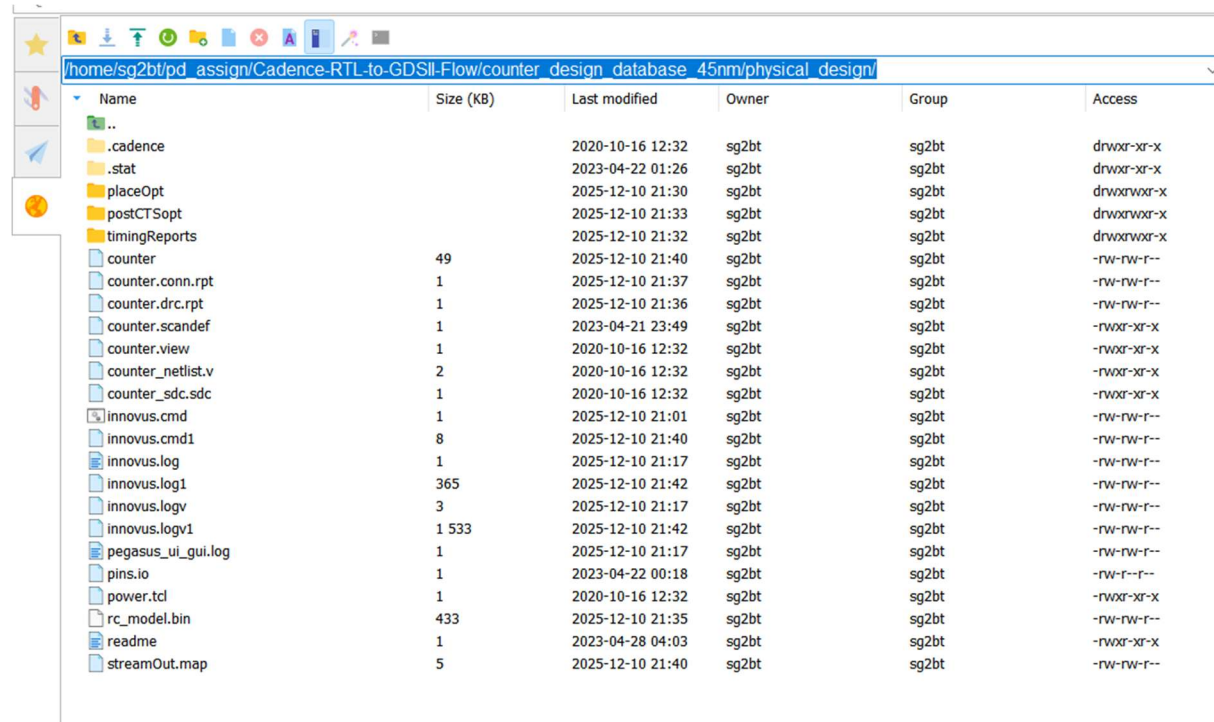
CTS and Routing Lab:

Below is the screenshot for CTS and routing stage that showing the connections inside the cell.



Generating GDS Stream File Lab:

Below is the screenshot of path for the GDS file.



Name	Size (KB)	Last modified	Owner	Group	Access
..					
.cadence		2020-10-16 12:32	sg2bt	sg2bt	drwxr-xr-x
.stat		2023-04-22 01:26	sg2bt	sg2bt	drwxr-xr-x
placeOpt		2025-12-10 21:30	sg2bt	sg2bt	drwxrwxr-x
postCTSopt		2025-12-10 21:33	sg2bt	sg2bt	drwxrwxr-x
timingReports		2025-12-10 21:32	sg2bt	sg2bt	drwxrwxr-x
counter	49	2025-12-10 21:40	sg2bt	sg2bt	-rw-rw-r--
counter.conn.rpt	1	2025-12-10 21:37	sg2bt	sg2bt	-rw-rw-r--
counter.drc.rpt	1	2025-12-10 21:36	sg2bt	sg2bt	-rw-rw-r--
counter.scandef	1	2023-04-21 23:49	sg2bt	sg2bt	-rwxr-xr-x
counter.view	1	2020-10-16 12:32	sg2bt	sg2bt	-rwxr-xr-x
counter_netlist.v	2	2020-10-16 12:32	sg2bt	sg2bt	-rwxr-xr-x
counter_sdc.sdc	1	2020-10-16 12:32	sg2bt	sg2bt	-rwxr-xr-x
innovus.cmd	1	2025-12-10 21:01	sg2bt	sg2bt	-rw-rw-r--
innovus.cmd1	8	2025-12-10 21:40	sg2bt	sg2bt	-rw-rw-r--
innovus.log	1	2025-12-10 21:17	sg2bt	sg2bt	-rw-rw-r--
innovus.log1	365	2025-12-10 21:42	sg2bt	sg2bt	-rw-rw-r--
innovus.logv	3	2025-12-10 21:17	sg2bt	sg2bt	-rw-rw-r--
innovus.logv1	1 533	2025-12-10 21:42	sg2bt	sg2bt	-rw-rw-r--
pegasus_ui_gui.log	1	2025-12-10 21:17	sg2bt	sg2bt	-rw-rw-r--
pins.io	1	2023-04-22 00:18	sg2bt	sg2bt	-rw-r--r--
power.tcl	1	2020-10-16 12:32	sg2bt	sg2bt	-rwxr-xr-x
rc_model.bin	433	2025-12-10 21:35	sg2bt	sg2bt	-rw-rw-r--
readme	1	2023-04-28 04:03	sg2bt	sg2bt	-rwxr-xr-x
streamOut.map	5	2025-12-10 21:40	sg2bt	sg2bt	-rw-rw-r--